

A Compact and High-Performance Eddy-Current Sensor Based on Meander-Spiral Coil

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In this paper, a novel meander-spiral coil (MSC) was designed as an eddy-current sensor (ECS) probe, which has small sensing area and spot size, nearly no limitation of mounting space, and can perform better than the regular-spiral coil (RSC). The results of the finite-element analysis show that the MSC has a magnetic active area that was only a quarter of that of the RSC. The MSC has much smaller inductance and resistance than the RSC, though the former has high sensitivity similar to the latter. These types of spiral coils were manufactured by printed circuit board technology and tested, which proves that the MSC has a small sensing range. Owing to MSC's small magnetic active area and high performance, this type of spiral coil is suitable for applications in sensor arrays and micro-electromechanical systems. Furthermore, using the MSC, ECSs have the potential to be used for micro-gap measurement, accelerometers, and pressure measurement applications, as in the case of capacitive sensors. Moreover, ECSs are inherently immune to environmental contaminations.

Index Terms—Displacement measurement, eddy currents, inductance, magnetic fields, sensor systems.

I. INTRODUCTION

APPLICATIONS of micro-displacement sensors [1] have rapidly grown along with the development of precision machinery and micro-electromechanical systems [2]. The most popular compact and high-resolution displacement sensors are capacitive and eddy-current sensors (ECSs) [2]. Capacitive sensors are characterized by a simple structure, high thermal stability, and extremely high resolution. Capacitive sensors have been widely used in all types of displacement, accelerometers, vibration and speed measurements, and pressure sensors. However, capacitive sensors are sensitive to the properties of dielectrics in the measuring gap, thus limiting their applications [3]. ECSs are usually low cost, high speed, inherently robust in harsh environments [4], and insensitive to contaminants [5], such as dirt, dust, oil, humidity, or any non-conductive material. Hence, ECSs have been widely used in various technical and industrial applications, such as displacement and vibration measurement, material property detection [6], crack fatigue inspection, and quality monitoring [7].

Sensing coil is the primary component of an ECS. The geometry parameters of the coil determine the effective measuring range, linearity, sensitivity, and the optimal working frequency of an ECS. The cost, stability, and working temperature range of an ECS probe depend on the coil manufacturing techniques. In the past, small ECS probes are primarily made up of wire-wound coils. The wire must be packed in perfect layers (usually by hand) under a microscope, and then inserted in a protective holder and potted with epoxy to stabilize the windings mechanically [5].

A wire-wound coil is a labor-intensive product (high cost), and the parameters and quality of the handmade coils are usually uncontrollable. With the development of different types of manufacturing processes, planar coils could be a better choice for eddy-current displacement sensors [8]–[11]. In general, planar coils have better thermal stability and better coil parameter consistency compared with other coils [12]. The printed circuit board (PCB) [13], thick film [14], electron beam or photo-lithography, and low temperature co-fired ceramic technologies [15] can all be employed to manufacture planar coils [16]. Through these technologies, high-quality ECS probes can be manufactured in bulk at an extremely low cost. An ECS with planar coil has several advantages, such as small spot size, low cost, high thermal stability, a compact structure, and high self-resonant frequency (SRF).

Given that the magnetic field of a sensor coil distributes inside the coil, as well as in a large region outside the coil, conductive material except the target entering into the field causes side loading [17]. The most common form of side loading comes from the mounting configuration, which reduces the sensor performance, and introduces significant measurement errors. The sensor coil target usually needs a size three times of the coil diameter because of the large magnetic field distribution area of the coil. An effective method to solve this problem is to use a magnetic shield shell for the sensor probe. However, the shield shell has poor temperature stability and large offset drift, which increases the complexity of the probe structure and packaging. Furthermore, ferromagnetic materials (e.g., shield shell materials) are unsuitable for some applications, such as magnetic resonance imaging system. A new method using two concentric coils carrying opposite currents [18] was proposed to reduce edge effect of eddy-current testing, which could be used to solve the side-loading effect of eddy-current displacement sensors. Compact ECSs without magnetic shielding would be very useful in displacement, vibration, and proximity measurement [19], as well as micro-gap measurement [20], accelerometer [21],

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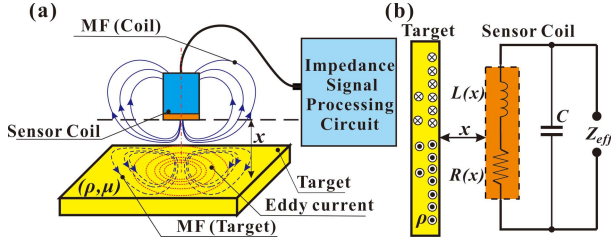


Fig. 1. (a) Working principle of ECSs. (b) Equivalent circuit of an ECS.

and pressure measurement applications [22], same as capacitive sensors. Moreover, ECSs are immune to environmental contaminations.

In this paper, a novel meander-spiral coil (MSC) was proposed as a compact ECS probe for micro-system and sensor-array applications. This planar coil is composed of several concentric spiral coils with different directions connected in series. Considering that the magnetic field lines of this MSC usually form closed curves from a sub-spiral coil to another, the active area of the sensor coil focuses on the nearby region of the coil. Thus, the proposed MSC can be perfectly used for all types of applications with limited mounting space. The magnetic field areas of the MSC and the regular-spiral coil (RSC) are calculated and compared through finite-element analysis (FEA). The performances of a group of planar coils with different wound structures that were manufactured by PCB technology were tested and compared. Both FEA and the experimental results show that the MSC achieves high sensitivity similar to the RSC, despite the MSC has smaller inductance and its magnetic field is distributed only in the surrounding area of the coil. The displacement response results show that the MSC has a small measuring range, but high sensitivity. Moreover, small inductance will increase the SRF, which makes the MSC can work at a high frequency, thereby enhancing sensitivity and stability.

II. PHYSICS AND COIL DESIGN

In general, a typical ECS system [23] consists of a sensor coil, a conductive target, and a signal conditioner board [Fig. 1(a)]. When driven by an ac current, the sensor coil generates alternating magnetic fields that induce eddy currents in the nearby conductive target. Likewise, the eddy currents generate magnetic fields, but in a direction opposite to that of the sensor coil. This phenomenon reduces the inductance of the sensor coil. The eddy currents dissipate energy as well, which increases the resistance of the coil. The moving target changes the coupling between the coil and the target, which is reflected in the variation of the impedance of the coil. The performance of an ECS can be quantified by the effective impedance [23] of the sensor coil corresponding to the target position x . Effect impedance is a complex function that depends on the characteristics of the sensor coil, target, and coil–target coupling. The function can be defined as follows:

$$Z_{eff} = R(x, \sigma, \mu, f) + j\omega L(x, \sigma, \mu, f) \quad (1)$$

where x is the target distance, σ and μ are the conductivity and permeability of the target, and f is the working frequency of the ECS. f , σ , and μ are all constants for a specific ECS with selected target and working frequency. The impedance of

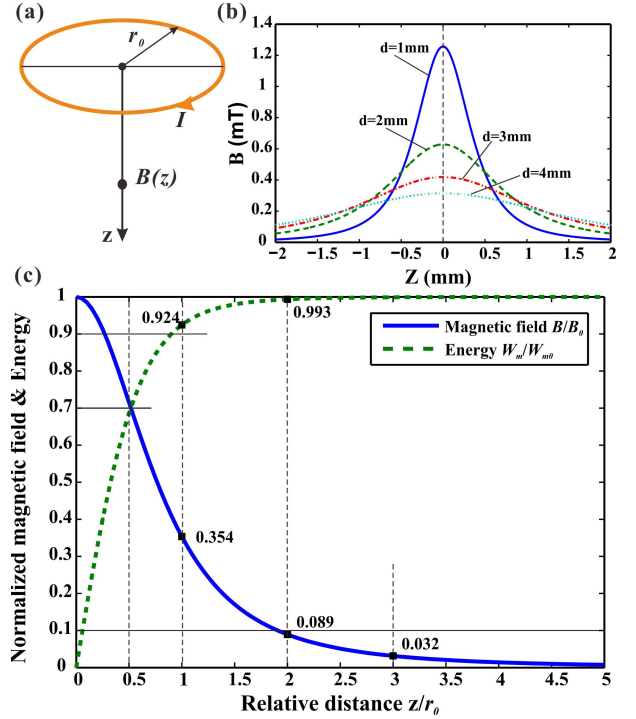


Fig. 2. (a) Ring loop coil carrying current. (b) Magnetic field density of the ring loop coils with different diameters. (c) Normalized magnetic field density and magnetic field energy.

the sensor coil can simply be represented as a series LR circuit and the parasitic capacitance [Fig. 1(b)]. Both the inductance L and the resistance R change with the target distance x , which can be expressed as follows:

$$Z_{eff} = R(x) + j\omega L(x). \quad (2)$$

In general, sensitivity and thermal stability increase along with working frequency and target conductivity. As reported in [24], ferromagnetic target is unsuitable for a high-performance ECS. The best target material is copper or aluminum. Silver is a slightly better option, but it is considerably more expensive than the others. The working frequency is limited by the SRF of the ECS probe. The SRF can be expressed as follows:

$$SRF = \frac{1}{2\pi\sqrt{LC}} \quad (3)$$

where L is the inductance of the unloaded coil (coil without target), which depends on the turns and the geometry of the coil, and C is the parasitic capacitance, including the capacitance of the sensor cable and the interwinding capacitance of the coil.

The sensitivity of an ECS depends on the magnetic field distribution of its coil. According to the Biot–Savart law, the magnetic field in the axis of a ring loop coil can be calculated by

$$B(z) = \frac{\mu_0 I r_0^2}{2(r_0^2 + z^2)^{3/2}} \quad (4)$$

where μ_0 is the permeability of vacuum, I is the total current flow through the coil loop, r_0 is the outer radius of the coil, and z is the distance to the coil center [Fig. 2(a)].

The magnetic field density in the axis of the coil with different diameters carrying 1 A current were calculated [Fig. 2(b)]. Evidently, the magnetic field density of the coil rapidly decreases with increasing distance z . The magnetic field distribution area depends on its diameter (radius). The smaller the coil diameter is, the higher the magnetic field near the coil is, and the faster the magnetic field decreases far from the coil surface [Fig. 2(b)]. To demonstrate the essential relationship between the magnetic field density and the coil diameter, the normalized magnetic field density $B(z)/B_0$ versus the z/r_0 can be calculated by

$$\bar{B}(z) = \frac{B(z)}{B(0)} = \frac{r_0^3}{(r_0^2 + z^2)^{3/2}} = \frac{1}{[1 + (z/r_0)^2]^{3/2}}. \quad (5)$$

The magnetic field density rapidly decreases with increasing distance z , which is 8.9% at $z = 2r_0$ [Fig. 2(c)]. According to the definition of magnetic field energy, the normalized magnetic field energy from the coil surface to z distance can be expressed as follows:

$$\begin{aligned} \bar{W}_m(z) &= \frac{W_m(z)}{W_m(\infty)} = \frac{\int_0^z \frac{B^2(z)}{2\mu_0} S \cdot dz}{\int_0^\infty \frac{B^2(z)}{2\mu_0} S \cdot dz} \\ &= \frac{16}{3\pi} \int_0^z \frac{1}{[1 + (z/r_0)^2]^{3/2}} dz \end{aligned} \quad (6)$$

where $W_m(z)$ is the total energy of the magnetic field from the coil surface to distance z and S is the equivalent magnetic area. 92.4% of the magnetic field energy of the coil is distributed between the coil surface ($z = 0$) and $z = r_0$, and nearly all the energy of the coil (99.3%) is concentrated in the area $z < 2r_0 = d$ [Fig. 2(c)]. Thus, the effective sensing range of a spiral coil is less than r_0 .

The structure of RSC, which is shown in Fig. 3(a), has a similar magnetic field distribution to the single-loop coil. Considering that magnetic field distribution depends on the coil diameter, an ECS should have high inductance sensitivity to target position x and high magnetic field energy. Therefore, a planar coil composed of a plurality of concentric spiral coils with different directions connected in series [Fig. 3(b)], namely, MSC, was designed for ECS to achieve high performances. With the help of FEA by COMSOL Multiphysics, the magnetic fields of two types of spiral coils were calculated with an axisymmetric model [Fig. 2(c) and (d)]. In the axisymmetric model, both coils have 20 turns of copper wire and an outer diameter of 30 mm. The copper wire has a rectangular cross section with a $152 \mu\text{m}$ width and $35 \mu\text{m}$ thickness. The vector arrow figures [Fig. 3(c) and (d)] illustrate that although the MSC is much complicated, it has a smaller magnetic field region compared with the RSC.

The magnetic field distribution and the resistance and inductance of the ECS system were analyzed through a frequency-domain study of the magnetic field (COMSOL Multiphysics 4.4b). The 2-D axisymmetric magnetic field interface in the frequency domain is selected where the following conditions are assumed:

$$(j\omega\sigma - \omega^2\epsilon)\mathbf{A} + \nabla \times (\mu^{-1}\mathbf{B}) = \mathbf{J}_e \quad (7)$$

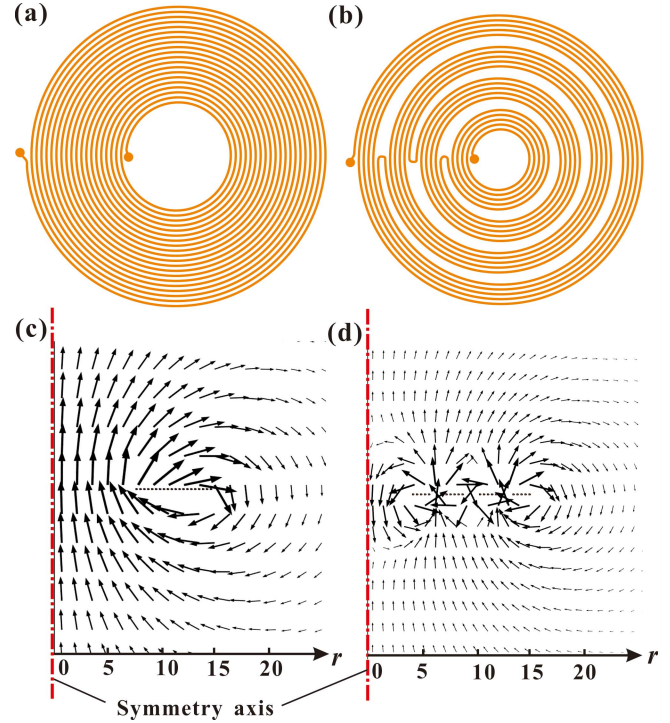


Fig. 3. Planar coils and their magnetic fields. (a) RSC. (b) MSC. Magnetic fields of (c) RSC and (d) MSC.

where ϵ denotes the permittivity, A is the magnetic vector potential, and J_e is the externally generated current density. A square-shaped air domain with large enough geometry was used to obtain accurate results during a reasonable computing time. Both sensor coils were excited by 1 MHz sine-wave currents and composed of copper wire, which has a conductivity of $5.998 \times 10^7 \text{ S/m}$. Aluminum targets were selected for the ECSs, which have a conductivity of $3.774 \times 10^7 \text{ S/m}$ and a thickness of 0.25 mm. Fig. 4(a) and (b) shows the magnetic field distributions of the MSC- and RSC-based ECSs with aluminum targets. The MSC-based ECS has a much smaller magnetic field region compared with the RSC-based ECS. The magnetic field lines of the two ECSs and the eddy currents in their targets are shown in Fig. 4(c) and (d), which clearly shows the different characteristics of the two ECSs. The FEA results indicate that the MSC-based ECS has a much smaller magnetic field area compared with RSC-based ECS. Thus, the MSC probe can be used as an ultra-compact eddy-current displacement sensor. The inductance and resistance of the two coils with different targets were calculated by FEA. Without a target, the RSC has an inductance of $11.19 \mu\text{H}$ and a resistance of 4.389Ω , whereas the MSC only has an inductance of $2.215 \mu\text{H}$ and a resistance of 3.699Ω . This can be attributed to the fact that the MSC counteracts part of the inductance of forward turns of the coil, resulting in lower total inductance. According to the traditional view of the ECS probe design, MSC is unsuitable for an ECS because it has a much smaller inductance than the RSC with a similar diameter. The sensitivity of the coil impedance to target displacement matters because it determines the feasible performances of the ECS.

To determine the sensitivity of the impedance to target displacement of the MSC and RSC, a parameter swept

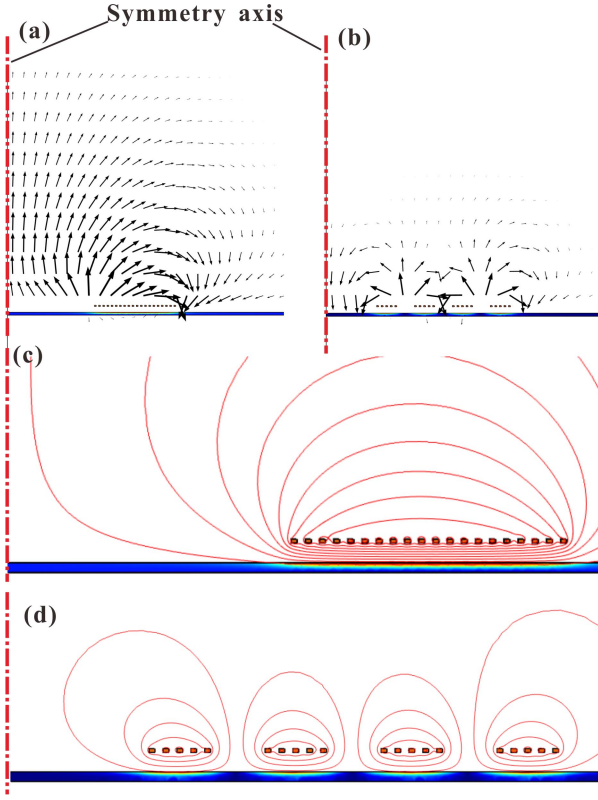


Fig. 4. Magnetic field and eddy-current distribution of planar coil-based ECSs. (a) Magnetic field of an RSC with target. (b) Magnetic field of an MSC with target. (c) Eddy-current density and magnetic field line of RSC with target. (d) Eddy-current density and magnetic field line of MSC with target.

analysis was conducted by COMSOL Multiphysics to obtain the impedance of the ECSs with these coils at different target distances. The results of the normalized impedance to the displacement were shown in Fig. 5(a). The curves show that both the resistance and the inductance of the MSC has similar responses to the displacement compared with the RSC, but the effective displacement measuring range of the MSC is much smaller than the RSC, which is derived from the small magnetic field distribution area of the MSC. Considering that inductance is commonly employed to interpret the target displacement, the following discussion involves only the relationship between the coil inductance and the target displacement. The ECS is sensitive to the target displacement before the normalized inductance reaches 0.9 (90% inductance of the unload coil), which can be considered to be the measuring range of the ECS. The MSC-based ECS has a measuring range of 2 mm ($0.13 r_0$), only a quarter of that of the RSC-based ECS (8 mm, $0.53 r_0$) [Fig. 5(a)].

The ECS performance depends on the relative sensitivity ($\bar{S}$) of the normalized inductance variation to the displacement change, which can be expressed as

$$\bar{S} = \frac{\Delta L/L_0}{\Delta x} = \frac{dL}{dx}/L_0. \quad (8)$$

The sensitivities of the impedance to the displacement of the coils have also been calculated using (8). The MSC has

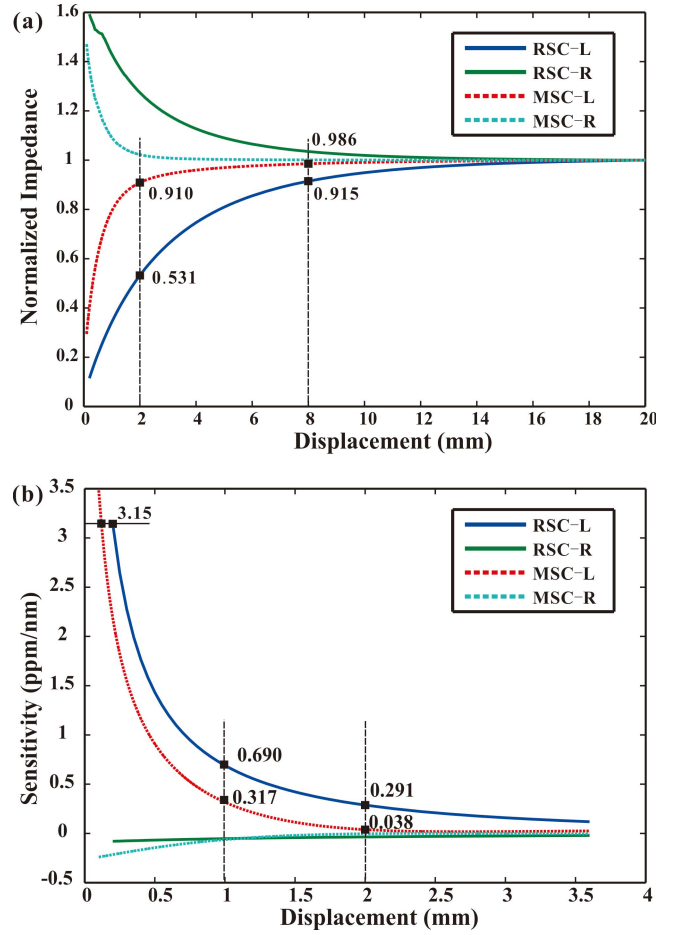


Fig. 5. FEA results of the response of the planar coil-based ECSs. (a) Normalized impedance to target displacement. (b) Impedance sensitivity to target displacement.

a nearly similar high sensitivity (3.15 ppm/nm) to the displacement compared with the RSC when the target distance is small, even though the inductance of the MSC is less than one-fifth of the RSC at a similar working frequency [Fig. 5(b)]. Thus, the MSC can be employed in eddy-current displacement sensors with high sensitivity. MSC sensitivity declined more rapidly than the RSC with increasing distance, which reduces to 0.038 ppm/nm at a 2 mm distance, only 1.2% of the sensitivity at zero distance.

Furthermore, the low inductance makes the MSC work at high frequency (high SRF), which means high sensitivity and stability and thin target. Hence, the eddy-current displacement sensor with an MSC could achieve high performance, even better than with an RSC. For a traditional ECS, the target must have a size three times of the sensor coil diameter. However, the MSC-based ECS could accurately measure the target position with a target dimension of less than 1.25 times sensor diameter, as the eddy currents in the targets reduce to less 0.1% of the maximum eddy current density at $r = 1.25r_0$. Overall, these characteristics contribute to the MSC-based ECS's suitability for high-performance displacement measurement in micro-electromechanical applications, particularly when the magnetic materials (magnetic shield) are unsuitable or the stability is concerned.

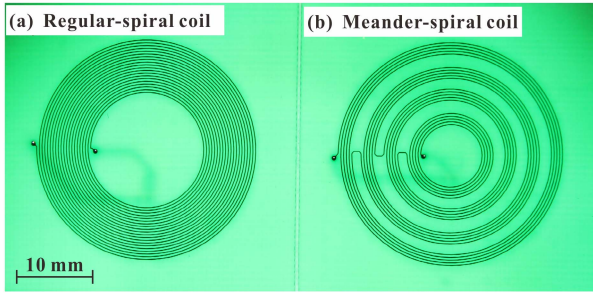


Fig. 6. Images of the planar coils by PCB technology. (a) RSC. (b) MSC.

III. EXPERIMENTAL RESULTS AND DISCUSSION

A group of single-layer planar coils were designed and manufactured by PCB technology. Each coil has an outer diameter of 30 mm with a $152\ \mu\text{m}$ width and $35\ \mu\text{m}$ thick copper wire. The pitch of the two wires in the coils is $0.36\ \text{mm}$. The coils were manufactured on a $40\ \text{mm} \times 40\ \text{mm}$ square board with a $1.6\ \text{mm}$ thick FR-4. First, the MSC and RSC with similar parameters to the FEA model were manufactured (Fig. 6). The experimental results are consistent with the simulation results by FEA. RSC has an inductance of $10.992\ \mu\text{H}$ and a resistance of $6.773\ \Omega$, whereas MSC has an inductance of $2.346\ \mu\text{H}$ and a resistance of $4.614\ \Omega$. The cable that joined the coil and the *LCR* meter has an inductance of $0.216\ \mu\text{H}$ and a resistance of $0.452\ \Omega$. The observed differences between the measured results and the FEA results are primarily derived from the parameter error of the PCB coil to the ideal model, particularly the thickness and width of the copper wire in the PCB coils, which contributed to the large resistance errors.

The impedance responses of the coils to the aluminum target displacement were measured by a precision *LCR* meter (LCR-8101, GW Instek Company Ltd., Taiwan) with a stable frame assembled a micro-positioning stage. The relationship between the coil impedance (resistance and inductance) and the target displacement are almost similar to the FEA results (Fig. 7). When the distance is small, the inductance sensitivity of the MSC can be even larger than the RSC, as high as over $3.5\ \text{ppm/nm}$. Therefore, MSC can distinguish a tiny variation of target distance as small as sub-nanometer. The resistance sensitivities of the coils are small. The resistance sensitivity of MSC is larger compared with that of the RSC. The curves show that the effective measuring range of the RSC can be as large as $8\ \text{mm}$ ($0.53\ r_0$), whereas less than $2\ \text{mm}$ ($0.13\ r_0$) for the MSC. Hence, the MSC-based ECS can achieve high performance, as well as be mounted in a limited space.

In the traditional view, ECS has a relatively large magnetic field active area and spot size in the target, which requires a few rules on the sensor mounting to maintain the performance. The nearest conductive material must keep a least distance to the sensor coil [Fig. 8(a)]. When the sensor coil has a diameter of d , a should be larger than d , and c should be over $3d$. Considering the eddy-current distribution in the target, the size of the target b should be larger than $3d$. Given the aforementioned characteristics, the mounting rules must be followed, which make ECS hardly used as a compact

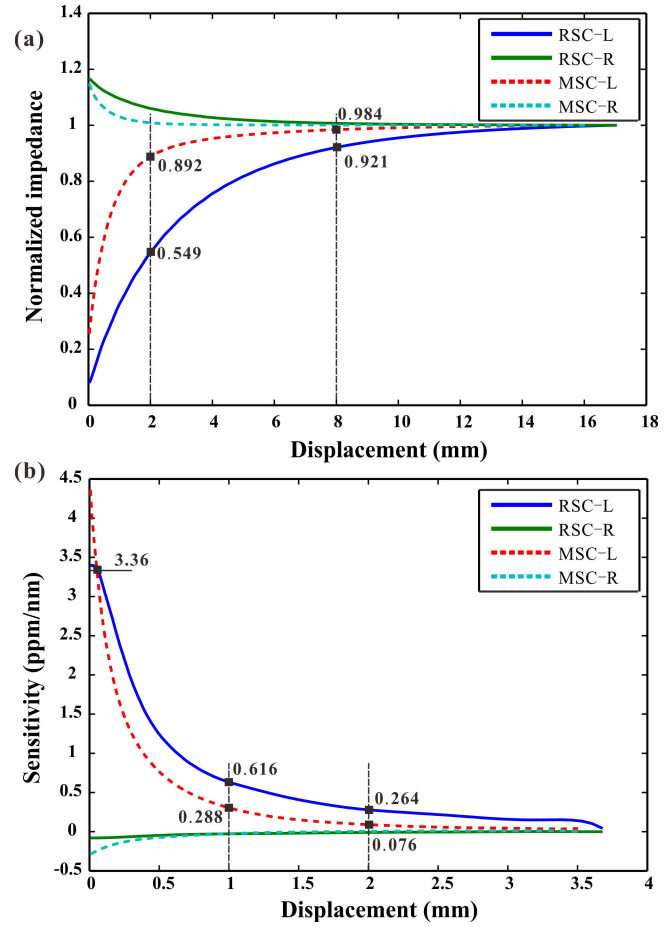


Fig. 7. Experimental results of the response of the planar coil-based ECSs. (a) Normalized impedance to target displacement. (b) Impedance sensitivity to target displacement.

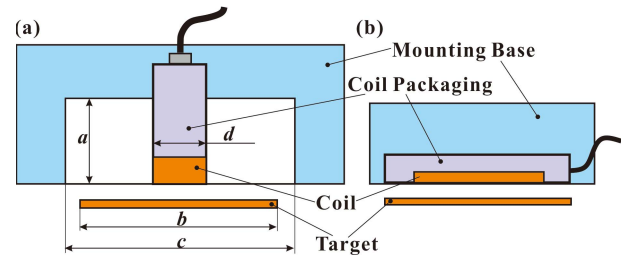


Fig. 8. (a) Space limitation of ECS mounting. (b) MSC-based micro-ECS system.

sensor when the mounting space is limited. The overlapping fields of closely mounted ECS probes will interfere and result in erroneous measurements when an array of ECSs was used. According to the FEA and experimental results, the sensing range of the MSC is only a quarter of that of the RSC with a similar diameter. The eddy-current distribution area in the targets is less than two times of the coil diameter for RSC-based ECS [Fig. 4(c)] and 1.25 times for MSC-based ECS [Fig. 4(d)], which means that the MSC probe has a small spot size of $1.25d$ (both b and c), comparable with capacitive sensors [3]. Furthermore, the sensing range of the MSC behind the coil could be less than $0.25d$ (one-fourth of that of the RSC), which includes 99% of the total magnetic

field energy, obtained by integrating the magnetic field energy. The sensing range of the MSC can be reduced or increased by increasing or decreasing the number and width of the sub-coils in the MSC. Such changes will consequently alter the magnetic field distribution. Such a small active area in the magnetic field could make the ECS probe insensitive to any conductive materials beyond the base board of the coil and the mounting shell. An MSC-based micro-ECS can be designed and manufactured for micro-mechanical system applications, which has nearly no limitations of mounting space and can achieve both high resolution and high thermal stability [Fig. 8(b)]. In addition, this compact ECS is perfect for sensor-array and integrated-electromechanical system applications. The signal conditioner and the MSC can be integrated and packed in a small probe, or even designed as an integrated circuit (IC) component.

IV. CONCLUSION

This paper presented a novel MSC as an ECS probe, which was designed by reversing a few turns of an RSC. The MSC has small sensing area and spot size, nearly unlimited mounting space, and can achieve better performance than RSC. The small sensing area of the MSC comes from the inner closed-form magnetic field line nearby the coil wire. The FEA results show that the MSC has a magnetic field active area that was only a quarter of that of the RSC. The MSC has much smaller inductance and resistance than the RSC. However, the MSC has high sensitivity similar to the RSC. These types of spiral coils were manufactured and tested. The experimental results show almost similar characteristics of both coils compared with the FEA results.

Given that the sensitivity range of the coil in the other side of the coil, far from the target, can be as small as $0.25d$ and the spot size in the target and the sensing area around the coil can be less than $1.25d$, the MSC is perfect for sensor-array and micro-electromechanical system applications. The MSC can be designed as an IC component for different technological applications. Furthermore, using this compact MSC probe, ECS can be used in displacement, vibration, and proximity measurement, as well as in micro-gap measurement, accelerometer, and pressure measurement applications, same as capacitive sensors, but inherently immune to environmental contaminations.

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